

Javascript

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Brief History of JavaScript

- JavaScript, often abbreviated as JS, was created by Brendan Eich in 1995 while he was working at Netscape Communications Corporation.
- JavaScript and Java are completely different languages, both in concept and design.
- Originally named "LiveScript," it was later renamed JavaScript to align with the popularity of Java.

What is JavaScript?

- JavaScript is a high-level, interpreted programming language primarily used for client-side web development, but it's also used in server-side development (Node.js), mobile app development, game development, and more.
- It is an essential component of modern web development, enabling dynamic and interactive web pages.

Basic Syntax and Structure

- JavaScript syntax is similar to other programming languages like Java and C, but it also has unique features specific to web development.
- JavaScript code is typically embedded within HTML using `<script>` tags or included as external script files.
- Statements are terminated with semicolons ;
- blocks of code are enclosed within curly braces { }

Javascript

- In HTML, JavaScript code is inserted between <script> and </script> tags.

```
<script>  
    document.getElementById("demo").innerHTML = "My First JavaScript";  
</script>
```

- Or, put the JavaScript in external files

```
<script src="myScript.js"></script>
```

<script> inside <head> and/or <body>

- Scripts can be placed in the <body>, or in the <head> section of an HTML page, or in both.
- Placing scripts at the bottom of the <body> element improves the display speed, because script interpretation slows down the display.

Output in JavaScript

- Writing into an HTML element, using `innerHTML`.
- Writing into the HTML output using `document.write()`.
- Writing into an alert box, using `window.alert()`.
- Writing into the browser console, using `console.log()`

JavaScript Identifier

- Identifiers are used to name variables and keywords, and functions.
- The rules for legal names are the same in most programming languages.
- A JavaScript name must begin with:
 - A letter (A-Z or a-z)
 - A dollar sign (\$)
 - Or an underscore (_)

JavaScript Identifier

- All JavaScript identifiers are case sensitive.
- The variables firstName and firstname are two different variables
- Combining multiple words into variable name.

Underscore:

first_name, last_name, master_card, inter_city.

Upper Camel Case (Pascal Case):

FirstName, LastName, MasterCard, InterCity.

Lower Camel Case:

JavaScript programmers tend to use camel case that starts with a lowercase letter:

firstName, lastName, masterCard, interCity.

JavaScript Variables

- Variables are Containers for Storing Data
- JavaScript Variables can be declared in 4 ways:
 - Automatically
 - Using **var**
 - Using **let**
 - Using **const**

Note

The **var** keyword was used in all JavaScript code from 1995 to 2015.

The **let** and **const** keywords were added to JavaScript in 2015.

The **var** keyword should only be used in code written for older browsers.

JavaScript Functions

- A JavaScript function is a block of code designed to perform a particular task.
- A JavaScript function is executed when "something" invokes it (calls it).

```
function name(parameter1, parameter2, parameter3)
{
    // code to be executed
}
```

- Function **parameters** are listed inside the parentheses () in the function definition.
- Function **arguments** are the **values** received by the function when it is invoked.

JavaScript Events

- HTML events are "**things**" that happen to HTML elements.
- When JavaScript is used in HTML pages, JavaScript can "**react**" on these events.

Event	Description
onchange	An HTML element has been changed
onclick	The user clicks an HTML element
onmouseover	The user moves the mouse over an HTML element
onmouseout	The user moves the mouse away from an HTML element
onkeydown	The user pushes a keyboard key
onload	The browser has finished loading the page

JavaScript Events Example

```
<button onclick="document.getElementById('demo').innerHTML = Date()">The time is?</button>
```

JavaScript Event Listener

- The `addEventListener()` method attaches an event handler to the specified element.
- Syntax: `element.addEventListener(event, function, useCapture);`
- Example:

```
element.addEventListener("click", myFunction);
```

```
function myFunction() {  
    alert ("Hello World!");  
}
```